Executive Summary

Leidos thanks the Centers for Medicare and Medicaid Services (CMS) and the Assistant Secretary for Technology Policy/Office of the National Coordinator for Health Information Technology (hereafter ASTP/ONC) for the opportunity to respond to the Request for Information (RFI) during the current public comment period.

As CMS pursues a bold agenda to enhance transparency, drive value-based outcomes, empower patients, expand accountable care, and modernize the U.S. healthcare ecosystem, it's clear thoughtful application of technology can be an essential enabler. From digital health tools that support chronic disease prevention to open, standards-based platforms that improve interoperability and transparency, today's innovations can address longstanding challenges in healthcare delivery.

To modernize healthcare, CMS should emphasize:

- Reducing administrative burden,
- Expanding access to actionable health data, and
- Fostering AI-driven insights to support clinical decision-making, fraud detection, and population health strategies.

These priorities closely align with Leidos' commitment to harnessing the power of smarter technologies and integrated teams to help achieve the shared goals of improved outcomes, better experiences, lower costs, and equitable access to care.

Leidos, a Fortune 500® company, leverages its extensive expertise in health technologies to deliver impactful, outcome-based solutions that address today's healthcare challenges. Recognized as a Top 10 Health IT provider, Leidos integrates engineering, science, and clinical insights to develop scalable solutions for hospitals, health systems, and biomedical organizations. Supporting every U.S. federal health agency, Leidos deploys advanced electronic health record (EHR) management, agile and DevOps-based practices, and innovative IT environments to enhance clinical, operational, and financial outcomes.

Leidos' response to this public comment RFI addresses the needs of Patients and Caregivers (PC), Providers (PR), and Technology Developers/Vendors (TD). We appreciate the chance to provide input and look forward to receiving feedback as part of an ongoing dialogue focused on health quality measures for providers, large-scale technology trends, and human-to-human interactions

PART B: Patients and Caregivers

1. Patient Needs

PC-1. What health management or care navigation apps would help you understand and manage your (or your loved ones) health needs, as well as the actions you should take?

We see a growing number of mobile apps that are designed to support patients in these efforts, including Blue Button 2.0 Connected Apps such as iBlueButton. Yet most of these apps still don't provide the depth of support and features that patients need as they move across healthcare settings. Specifically, they lack interoperability between patients' healthcare providers, health systems, and other stakeholders.

We understand CMS' long-term vision of interoperability and urge the agency to continue to prioritize the patient digital health experience by supporting health policies and technology solutions that foster the growth of interoperability.

What's needed are applications that put interoperability and date exchange front and center—and bring about an enhanced digital experience for the beneficiary. Compliance with FHIR standards forms the foundation to allow seamless data sharing across providers, payers, and public health system. On top of that, mobile applications should have the ability to aggregate data from EHRs, claims, pharmacy, and social services.

Patients should be able to share personal health information with any healthcare stakeholder at any time, without fear that information from one physician on a different network will go missing—or prove difficult to obtain. Patients using consumer healthcare apps to track their heart rate or blood sugar levels, or send messages to physicians should feel confident that they won't have to repeat themselves as they move across the care continuum. We also encourage CMS to continue to support the development of digital health solutions (like consumer-facing apps) and other technologies including telehealth, messaging, scheduling, and patient portals.

To stimulate the growth of digital health, CMS should consider ways to incentivize its partners to develop and roll out additional, robust features that can be incorporated into patient-facing communications, such as AI voice-enabled technologies, to optimize return on investment.

PC-2. Do you have easy access to your own and all your loved ones' health information in one location (for example, in a single patient portal or another software system)?

Accessing health information is often challenging for patients and their caregivers, primarily due to a lack of interoperability.

CMS beneficiaries would benefit from apps that synthesize medical and clinical data (including historical health information) so it's accessible at a moment's notice. The challenge is that even the most comprehensive apps do not have the ability to share information across the continuum of care or display healthcare data in a caregiver-friendly format.

As CMS works to improve patient access, the agency must also focus on improving caregivers' experiences (e.g., spouses of patients with a primary diagnosis that requires round-the-clock care). Caregivers need to have easy access to health information at a moment's notice.

We suggest CMS and its partners continue strengthening digital health workflows that enable the transmission and receipt of critical, time-sensitive information to caregivers as needed (e.g., when a physician needs to determine whether to prescribe an opiate for pain for a patient with a history of substance-use disorders).

We also see the benefit of CMS adopting smarter enterprise imaging solutions, which are built with sophisticated sharing capabilities. Currently, many legacy imaging solutions are inefficient and lack advanced data-sharing features and require hundreds of thousands of dollars to maintain. CMS should consider upgrading cloud-based (imaging as a service) solutions, and/or partnering with a third-party provider that can leverage more sophisticated image-capture and image-share capabilities.

PC-3. Are you aware of health management, care navigation, or personal health record apps that would be useful to Medicare beneficiaries and their caregivers?

We are aware that a growing number of patients are using digital health—including 30% of adults over age 65. Yet while many of these apps are useful to caregivers—such as apps that help patients manage and adhere to medication schedules—the vast majority could use improvement.

Due to the lack of interoperability and incentives to engage patients and caregivers, adoption of digital health apps by patients and providers is still limited.

We recommend CMS work closely with its technology partners to roll out patient-centered digital health solutions or encourage the development of digital products that would have a measurable, meaningful impact on clinical outcomes. For example, utilizing the principle

of gamification by tying in some kind of measurable "reward"—such as points toward a tangible award for completing actions like checking blood pressure daily—could stimulate patient engagement.

PC-5. What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?

CMS should focus on actively marketing the benefits of digital health products directly to healthcare providers and consumers by extending its outreach efforts. In turn, CMS should support expansion of Medicare coverage of digital health solutions. These efforts have proven successful in the recent past.

Case in point: when CMS initially expanded telehealth coverage in March 2020 beyond rural territories, thousands of patients received lifesaving care they otherwise wouldn't have been able to access—which prevented thousands of adverse events. Today, as a result of these telehealth policy expansions, virtual care continues to benefit a greater number of Medicare and Medicaid beneficiaries who can now manage chronic conditions more effectively and avoid hospitalizations.

CMS beneficiaries shouldn't have to worry about steep out-of-pocket costs. Given ongoing physician shortages in most of the U.S., we should continue to grow digital health solutions that offer a clear, proven return on investment.

Swift action now to promote use of smart, digital health technologies will pay dividends far into the future.

In addition to supporting the expansion of coverage to digital health, and amplifying marketing efforts to patients and providers, CMS should continue to champion health policies and technology innovations that further the growth of interoperability.

PC-6. What features are most important to make digital health products accessible and easy to use for Medicare beneficiaries and caregivers, particularly those with limited prior experience using digital tools and services?

Putting the patient experience front and center is the key to making digital health products accessible and easy for Medicare beneficiaries. As third-party developers roll out digital health products and services—and as CMS weighs coverage decisions aligned with digital health solutions— it's important to consider:

- **Will this product or service make it easier for Medicare beneficiaries to maintain or improve key health metrics?** For example: while text-based notifications are often touted as a solution for all, developers of notification services need to look closely at studies to align specific actions (improvements in

medication adherence) with the frequency of notifications. If there is a clear correlation between daily reminders at a pre-determined time and outcomes improvement, that would demonstrate a particular messaging solution shows promise.

- **How costly would it be to implement a digital health solution?** Given the surge in overhead costs across all healthcare domains, CMS must be careful in its cost-benefit analysis.
- **Is a digital health solution, such as an app, customizable/configurable and scalable?** As population health needs evolve, working with third-party partners that can scale and adapt as regulations and needs shift is critical to elevating population health.

PC-7. If CMS were to collect real-world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so?

CMS would benefit from leveraging predictive analytics and AI platforms that can collect and read data to isolate clinical and financial trends. Combining the use of these technologies with HIE data and other information sources can help CMS gain detailed, in-depth insights into healthcare outcomes and related costs that would inform decision making and improve data interoperability.

2. Data Access and Integration

PC-9. Given that the Blue Button 2.0 API only includes basic patient demographic, Medicare coverage, and claims data (Part A, B, D), what additional CMS data sources do developers view as most valuable for inclusion in the API to enable more useful digital products for patients and caretakers?

We support the expansion of Blue Button 2.0 to provide Medicare Advantage (MA) beneficiaries Part D formulary content, which will require updating the underlying data schema and ensuring API support for plan tiering, utilization management indicators, and clinical detail.

We encourage CMS to work closely with MA and Part D sponsors to normalize and validate shared data structures. Our extensive work with Medicare Advantage and Medicaid encounter data processing can support CMS in these goals.

PC-10. How is the Trusted Exchange Framework and Common Agreement™ (TEFCA™) currently helping to advance patient access to health information in the real world?

Since going live in December 2023, TEFCA has gone from a great idea to a promising real-world framework that facilitates nationwide health information exchange among designated Qualified Health Information Networks (QHINs). We are encouraged to see use of the TEFCA exchange accelerating, with the retrieval of more than nine million documents and counting, according to recent estimates from The Sequoia Project.

We suggest that CMS continue to support TEFCA and incentivize network participation and work with partners committed to building out interoperability frameworks, such as Leidos.

PC-11. How are health information exchanges (HIEs) currently helping to advance patient access to health information in the real world?

User adoption of HIEs has not reached its full potential. Even in states like New York and Vermont, which have higher success rates with HIE adoption, onboarding providers remains challenging. As studies like this have pointed out, there is a notable gap between hospitals' adoption and the actual use of different HIE methods to send inside information to outside organizations, largely due to usage barriers (e.g., lack of training, inadequate funding, lack of infrastructure, privacy concerns, etc.).

Additional funding, support, and training, therefore, will be necessary to sustain the momentum around HIEs. We suggest creating additional incentives to providers, perhaps through vehicles such as the Merit-Based Incentive Payment System (MIPS) program, Accountable Care Organizations (ACO), technical assistance grants, or HIT modernization funding, to encourage participation from smaller and/or rural providers to connect to HIEs.

3. Information Blocking and Digital Identity

PC-13. How can CMS encourage patients and caregivers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would increasing reporting of complaints advance or negatively impact data exchange?

The best way to improve user engagement is by raising awareness of the issue of information blocking—and its consequences. The process to submit complaints must use clear examples and plain language for non-technical users. We also recommend the use of Sentiment Analysis technologies on submitted complaints to more effectively categorize, summarize and identify focus areas and trends.

PART C: PROVIDERS

1. Digital Health Apps

PR-1. What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved (see description in PC-5) digital health products for their patients?

While information-sharing technologies have evolved considerably, thus encouraging provider adoption of digital health apps, there are still a few obstacles CMS needs to address to boost provider engagement.

Challenges such as patient-record infrastructure and connectivity limitations, financial constraints, reimbursement, and the pressure to do more with less plague providers in all healthcare settings. Other challenges, such as information blocking, as well as a lack of interoperability, continue to stymie adoption of all digital health products.

With projected physician shortages of 40,800 to 120,000 by 2030, according to the Association of American Medical Colleges (AAMC), now is the time to double down on growing digital health solutions that work.

CMS should continue its efforts to foster innovation within the health technology ecosystem, enabling organizations to create digital health products for providers to care for their patients. The agency must continue to support interoperability, to protect the continuity of care: Providers should know that patient-encounter data is visible at every touchpoint of their experience, across the care continuum. Every time a clinician checks a patient's vitals, that data should become immediately available from that point forward, whether the patient's next visit is in the same facility or a different state.

It's impossible to convince patients to use digital health products without fully embracing and promoting interoperability.

Incentives also play a crucial role in generating physician buy in to any initiative, as we've seen since the onset of the EHR Incentive Program. In tandem with rolling out healthcare policies that incentivize providers for adopting and implementing technologies, CMS could work directly with third-party vendors to develop systems and solutions that demonstrate clear clinical and financial benefits.

PR-2. What are obstacles that prevent development, deployment, or effective utilization of the most useful and innovative applications for physician workflows, such as quality measurement reporting, clinical documentation, and billing tasks? How could these obstacles be mitigated?

The key barriers to development, deployment, and utilization of digital health applications include cost, accessibility, and provider adoption. We believe CMS should address provider challenges and risks through various mechanisms, such as a formal mentoring program, award recognition programs, annual achievement awards, training programs, structured lines of communication, and online portals.

Under the National Healthcare Safety Network (NHSN), Leidos has created a variety of technology solutions that meet the provider where they are to facilitate a variety of data delivery and sharing mechanisms.

PR-3. How important is it for healthcare delivery and interoperability in urban and rural areas that all data in an EHR system be accessible for exchange, regardless of storage format (for example, scanned documents, faxed records, lab results, free text notes, structured data fields)?

Ensuring that providers have the necessary data to accurately diagnose and improve the quality of patient care, regardless of format or location, remains paramount.

Integration allows data in an EHR to be accessible. While HIEs can facilitate integration between providers, getting disparate systems to communicate efficiently can prove challenging.

We support CMS in promoting standards for communicating within the healthcare environment, such as using Digital Imaging and Communications in Medicine (DICOM) standards for the movement and storage of medical imaging data, or Clinical Document Architecture (CDA) for document-based standards for clinical summaries. Even with the use of standards, providers must plan to enhance their security measures to protect patients' Personally Identifiable Information (PII), ensuring only those required or consented to access the data can do so.

PR-4. What changes or improvements to standards or policies might be needed for patients' third-party digital products to have access to administrative workflows, such as auto-populating intake forms, viewing provider information and schedules, and making and modifying an appointment?

CMS needs to push for an entirely digital, seamless patient-provider experience across the healthcare continuum. This includes everything from patient intake forms to lab tests and results, to data-rich image files.

Organizations must update changes that focus on API access, data availability, consent, and governance. Securing patients' PHI/PII information remains the utmost priority when allowing third-party access to pertinent information.

We recommend adopting a Zero Trust approach for capabilities such as authentication for every application and user, micro-segmenting APIs to control access to information, and using explicit user consent for data access with full audit and reporting.

2. Data Exchange

PR-5. Which of the following FHIR APIs and capabilities do you already support or utilize in your provider organization's systems, directly or through an intermediary? For each, describe the transaction model, use case, whether you use individual queries or bulk transactions, and any constraints:

We are particularly impressed with CMS's leadership in adopting and mandating FHIR standards within its health IT environment. This commitment to open standards is crucial for fostering a more connected and patient-centric healthcare system. We strongly believe that FHIR is the key to unlocking the full potential of healthcare data and enabling true interoperability.

We encourage CMS to continue prioritizing FHIR in future procurements and mandates. By further expanding the adoption of FHIR, CMS can drive even greater innovation, reduce provider burden, and improve the quality of care for all Americans.

PR-6. Is TEFCA currently helping to advance provider access to health information?

Yes, TEFCA is advancing provider access to health information for healthcare entities to support public health, quality reporting, and data exchange for smaller providers.

However, while TEFCA lays groundwork for nationwide exchange, it currently lacks support for real-time population-level FHIR data extraction. To help alleviate this challenge, CMS could incentivize adoption of the Bulk Data Access (Flat FHIR) standard for risk adjustment and research-grade analytics. CMS should also acknowledge the limitations of a

centralized TEFCA-only model for data exchange. Many niche, rural, or specialized providers may be unable to meet TEFCA's participation requirements or maintain the technical infrastructure required. CMS could offer hybrid or federated participation models where local networks retain autonomy but align with core exchange standards.

PR-7. What strategies can CMS implement to support providers in making high-quality, timely, and comprehensive healthcare data available for interoperability in the digital product ecosystem? How can the burden of increasing data availability and sharing be mitigated for providers? Are there ways that workflows or metrics that providers are already motivated to optimize for that could be reused for, or combined with, efforts needed to support interoperability?

We agree that providers, especially small and medium-size practices, need extra support when it comes to promoting interoperability and making healthcare data more readily available.

We recommend CMS do more to incentivize and standardize data quality and timeliness, such as aligning existing incentive programs (e.g., MIPS) to performance goals tied to interoperability. CMS should continue to work toward implementing a national framework for patient matching to reduce errors and improve data accuracy. In turn, CMS could work with healthcare industry stakeholders to identify workflows that providers are already motivated to optimize (e.g., documentation for billing and quality reporting).

PR-8. What are ways CMS or partners can help with simplifying clinical quality data responsibilities of providers?

We recommend the following strategies to simplify clinical quality data responsibilities:

1. **Standardize and Streamline Data Collection:** Continue to develop and promote a core set of standardized quality measures across all reporting programs (e.g., MIPS, ACOs).
2. **Automate Data Extraction and Reporting:** Expand the use of APIs (FHIR) to enable seamless data exchange between EHRs, registries, and CMS systems.
3. **Reduce Redundant Reporting:** Implement a "collect once, use many times" approach to data management by allowing providers to submit data once and use it for multiple reporting programs.
4. **Focus on Patient-Centered Outcomes:** Shift the focus from process measures to outcome measures that are more meaningful to patients.
5. **Collaborate with Stakeholders:** Engage with providers, EHR vendors, and other stakeholders to identify and address the challenges of quality reporting.

3. Digital Identity

PR-9. How might CMS encourage providers to accept digital identity credentials (for example, CLEAR, ID.me, Login.gov) from patients and their partners instead of proprietary logins that need to be tracked for each provider relationship?

We recommend a multi-pronged approach to encourage providers to use publicly acceptable digital identity credentials. CMS should focus on the creation of a secure, standards-based digital identity framework that is based on NIST 800-63-3 (IAL2/AAL2) tied together with OAuth 2.0, and OpenID Connect. This standardized approach to digital identity across CMS programs could improve interoperability by enabling consistent, authorized access to patient data across different platforms and services, facilitating better continuity of care and compliance with privacy regulations such as HIPAA.

This approach includes updating standards and certifications to align with ONC Health IT Certification Criteria, which would require EHR vendors to support OAuth 2.0/OpenID Connect. This update will enable patients to use federal digital identity providers, such as Login.gov, that enable patients to access their own health information.

To encourage providers to accept identify credentials, CMS will need to communicate their benefits. This would include showing that fewer manual processes related to account creation, identity verification, and password resets would be needed within the provider environment. Identity credentials also minimize the need to manage multiple proprietary login systems, reducing overhead and improving workflow efficiency. When patients can securely authenticate themselves using a trusted digital credential, providers gain faster access to accurate intake forms, insurance data, and consent records that then allow providers to focus on clinical care rather than administrative validation.

PR-11. How could members of trust communities [3] (for example, QHINs, participants and subparticipants in TEFCA, which requires Identity Assurance Level 2 (IAL2) via Credential Service Providers (CSPs)) better support the goals of reduced provider and patient burden while also enhancing identity management and security?

Trust communities should support federated identity models that allow for reusable, verified credentials across networks. This effectively enhances security and user convenience, and interoperability. Credential Service Providers (CSPs) meeting NIST IAL2 standards, such as Login.gov or ID.me, enable providers and patients to securely access multiple services without maintaining separate logins for each system, reducing friction and eliminating redundant identity verification across organizations.

Implementing multi-modal identity proofing options, such as biometric verification, ensures equitable access by accommodating varying levels of digital literacy and access to technology.

Enhancing security through role- and context-based access control limits data access based on a user's role and purpose, reducing unnecessary exposure to sensitive health information. Coupling this with transparent consent management tools allows patients and providers to easily manage access rights.

Aligning identity federation policies across QHINs and participants ensures consistency and predictability in managing identity and data exchange across the TEFCA ecosystem.

4. Information Blocking

PR-12. Should ASTP/ONC consider removing or revising any of the information blocking exceptions or conditions within the exceptions (45 CFR part 171, subparts B through D) to further the access, exchange, and use of electronic health information (EHI) and to promote market competition?

We recommend a targeted review of information blocking, which focuses on technology capability and resource availability. CMS might focus on areas where current regulations may hinder interoperability and innovation across the healthcare landscape. One method that CMS can support for encouraging interoperability is to grade facilities on their technology use and employment maturity.

PART E: TECHNOLOGY VENDORS, DATA PROVIDERS, AND NETWORKS

1.0 Ecosystem

TD–1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

We recommend expanding the use of innovation challenges and grant programs, modelled after ONC's past work, to incentivize solutions for chronic disease management, dementia care coordination, and other high-impact areas. CMS should expand funding for pilot programs and grants that reward early adoption of open, standards-based APIs such as FHIR and USCDI version 4. Targeted grants for small- and medium-size health IT developers addressing Medicare populations with high needs — such as dual eligibles or chronic disease management — can stimulate development of tailored, practical solutions. We envision CMS publishing a "wish list" of application and/or product needs in a series of prioritized challenges that align with those needs, thereby stimulating meaningful third-

party development and creating alignment between real-world beneficiary needs and developer solutions.

Another recommendation is to evolve the existing CMS developer sandbox to include richer, more dynamic synthetic datasets, and to provide simulated authentication, consent, and access scenarios. This would better reflect the real-world complexities developers face when creating apps for Medicare beneficiaries. CMS could further advance these efforts by expanding real-world testing feedback loops to better integrate with CMS performance measurement programs (e.g., electronic clinical quality measures, or eCQMs) which offer developers assurance their products will perform reliably in Medicare workflows. ASTP/ONC live demonstration tests both during initial testing and post certification in the real world that validate consistent API behavior and data quality are critical to reducing implementation risks and lowering the incidence of security breaches or incomplete data exchange.

TD-2. Regarding CMS Data, to stimulate developer interest-- a. What additional data would be most valuable if made available through CMS APIs? b. What data sources are most valuable alongside the data available through the Blue Button 2.0 API? c. What obstacles prevent accessing these data sources today? d. What other APIs should CMS and ASTP/ONC consider including in program policies to unleash innovation and support patients and providers?

We recommend that CMS continue building on its efforts to foster innovation across the health technology ecosystem by enhancing current initiatives such as Blue Button 2.0, the Synthetic Data Sandbox, and the Medicare FHIR APIs. While CMS has made significant progress in opening APIs and supporting developer access to synthetic Medicare data for purposes of increasing understanding of Medicare hosted data and improving potential data utility, targeted innovation is still needed to improve the efficiency and efficacy of care delivery and support policy and patient education.

CMS has made strides in FHIR API enablement, but current data types available through Blue Button 2.0 remain limited compared with the full range of Medicare data. Expanding access to include Part C encounter data and real-time pharmacy benefits—and aligning those data with population demographics and behavioral health indicators—would significantly improve the ability of third-party developers to design tools that improve care and outcomes.

Finally, CMS should consider forming strategic data partnerships to integrate population demographics and other cross-agency datasets using FHIR R4+ APIs. CMS has taken important steps to make data available, but many developers continue to face burdens in accessing and integrating this data for analytic and clinical applications. Developer

challenges include fragmented data governance—multiple agencies and contractors maintaining control over different data segments, leading to inconsistent access policies—and patient matching and data standardization issues, which hinder identity resolution and impede seamless integration. CMS can help developers overcome these challenges by promoting APIs such as Prior Authorization APIs leveraging HL7 Da Vinci Prior Authorization Support (PAS) standards to improve transparency and administrative efficiency, and Coverage Requirements Discovery (CRD) APIs to provide clarity on benefit coverage and reduce workflow complexity.

2.0 Digital Identity

TD–3. Regarding digital identity implementation: a. What are the challenges and benefits? b. How would requiring digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800–63–3 IAL2/AAL2 CSPs) impact cybersecurity and data exchange? cybersecurity and data exchange? c. What impact would mandatory use of the OpenID Connect identity protocol have?

We support CMS' commitment to secure equitable digital identity frameworks and recommend several enhancements to existing efforts. CMS already relies on federally approved identity providers like Login.gov and supports OAuth 2.0/OpenID Connect standards through its FHIR-based APIs. However, further steps are needed to broaden adoption, reduce beneficiary burden, and strengthen the digital identity ecosystem across the Medicare enterprise.

Leidos recommends that CMS expand support for NIST 800-63-3 IAL2/AAL2-compliant identity proofing by developing toolkits and libraries for developers, subsidizing identity verification for low-income beneficiaries, and requiring at least one certified credentialing integration for all CMS API consumers. However, mandatory compliance requirements must be carefully balanced to avoid marginalizing populations with limited internet access or digital skills. CMS and ASTP/ONC should support subsidized identity verification for low-income beneficiaries and provide fallback options, ensuring equity and inclusion aligned with FAVES principles—fairness, appropriateness, validity, effectiveness, and safety.

Equally important, CMS should continue to promote OAuth 2.0 and OpenID Connect as the baseline standards for authentication and token-based access. These technologies, when implemented through federated models, reduce credential fatigue for users and minimize attack surfaces. CMS could advance adoption by establishing a FHIR-based identity token exchange architecture across its digital properties and encouraging integration with state Medicaid systems and third-party app developers.

A promising next step for CMS would be the creation of a federated identity trust framework—a policy and technical architecture that allows CMS and its partners to recognize and validate credentials issued across trusted networks. This model has seen success in the federal research and education sector through platforms like InCommon, and it could be adapted for health use cases. CMS could pilot such an initiative by aligning with the ONC FAST Identity Tiger Team's recommendations and leveraging TEFCA participants as early adopters.

Finally, CMS should prioritize user-centered design when implementing digital identity solutions, especially for populations with low digital literacy. This includes supporting credential recovery, multilingual accessibility, and biometric alternatives. These human-centered methods would translate well to CMS initiatives involving seniors and underserved populations.

3.0 Technical Standards and Certification

TD-4. How can CMS better encourage use of open, standards-based, publicly available APIs over proprietary APIs?

CMS should promote API-first development by tying financial incentives, such as participation in value-based programs or EHR certification benefits, to the use of open, standards-based APIs like HL7 FHIR. Additionally, CMS can require certified systems to demonstrate compatibility with public implementation guides and discourage proprietary extensions that limit interoperability. To support adoption, CMS should also fund API toolkits, developer sandboxes, and public directories of certified APIs to lower the barrier for integration and innovation across the ecosystem.

TD-5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?

A nationwide provider directory of FHIR endpoints would significantly enhance access to health information by enabling patients, providers, and payers to discover and connect with data sources more efficiently, reducing friction in care coordination and data exchange. CMS should lead the publication and maintenance of this directory as a public good, ensuring it includes essential metadata such as supported data types, contact information, and identity assurance levels. This directory should include essential metadata such as validated National Provider Identifiers (NPI), organizational affiliations, supported data types, endpoint purposes, and contact information.

While CMS has taken initial steps through initiatives like the National Plan and Provider Enumeration System (NPPES) and endpoint registry pilots, broader standardization,

mandatory reporting, and real-time updates are needed. The directory should be freely accessible to ensure equitable use across stakeholders and promote adoption of interoperable APIs.

TD-6. What unique interoperability functions does TEFCA perform? a. What existing alternatives should be considered? b. Are there redundant standards, protocols or channels or both that should be consolidated?

TEFCA uniquely enables a nationwide, policy-based framework for health information exchange across disparate networks, offering standardized governance and minimum required capabilities for QHINs. CMS should consider aligning its interoperability programs with TEFCA's trust framework to reduce fragmentation. However, CMS should also assess existing alternatives—such as Carequality, CommonWell, and DirectTrust—that may offer more nimble or specialized exchange options for certain providers. To streamline interoperability, CMS should identify and consolidate redundant standards and protocols, particularly where overlapping FHIR-based APIs, Direct protocols, or regional HIE channels create unnecessary complexity for data exchange participants.

CMS should also acknowledge the limitations of a centralized TEFCA-only model for data exchange. Many niche, rural, or specialized providers may be unable to meet TEFCA's participation requirements or maintain the technical infrastructure required. CMS could offer hybrid or federated participation models where local networks retain autonomy but align with core exchange standards.

TD-7. To what degree has USCDI improved interoperability and exchange and what are its limitations?

USCDI has improved interoperability by providing a standardized structure for core clinical data, fostering consistency across EHR systems and APIs. However, its scope remains limited in supporting comprehensive, person-centered care. CMS should support the continued expansion of USCDI to prioritize data classes relevant to home health, long-term care, social determinants of health, and patient-reported outcomes. To better integrate unstructured and semi-structured data, CMS should also promote the use of flexible formats—such as HL7 CDA with embedded narrative text—that support natural language processing (NLP) and enhance the utility of less structured clinical content.

TD-8. What are the most effective certification criteria and standards under the ONC Health IT Certification Program?

We support CMS' goal to strengthen certification programs and promote technical standardization, and we recommend several enhancements to further advance these efforts. Among the most effective certification criteria under the ONC Health IT Certification Program are requirements for standardized APIs, use of USCDI, and support

for HL7 FHIR for patient data access and exchange. These have significantly advanced interoperability. However, to further strengthen the program, CMS and ONC should expand FHIR certification requirements to cover full-spectrum clinical and administrative data—including narrative notes, lab results, scanned documents, and structured demographic and population health data. Certification should also prioritize real-world data exchange performance and API response completeness to ensure that certified systems meet the functional needs of patients, providers, and payers.

TD-9. Regarding certification of health IT: a. What are the benefits of redefining certification to prioritize API-enabled capabilities over software functionality? b. What would be the drawbacks? c. How could ASTP/ONC revise health IT certification criteria to require APIs to consistently support exchanging data from all aspects of the patient's chart (for example, faxed records, free text, discrete data)? d. What policy changes could CMS make so providers are motivated to respond to API-based data requests with best possible coverage and quality of data? e. How could EHRs capable of bulk data transfer be used to reduce the burden on providers for reporting quality performance data to CMS? What capabilities are needed to show benefit? What concerns are there with this approach?

CMS should recommend that ONC revise health IT certification criteria to prioritize API-enabled interoperability over legacy software-centric functionality, focusing on real-time, standards-based data exchange. CMS and ASTP/ONC should collaborate to ensure that certification criteria prioritize transparency, data provenance, and record reconciliation, aligning with Health Data, Technology, and Interoperability (HTI) HTI-1, HTI-2 and HTI-3 final rules. Certification should require APIs to support access to all elements of the patient chart, including scanned documents, free text, and structured data, using consistent formats like FHIR and CDA. To motivate providers, CMS could tie reimbursement incentives or quality measures to the completeness and timeliness of API-based responses. Bulk FHIR capabilities should be leveraged to automate quality reporting and reduce provider burden, provided appropriate safeguards are in place to address data privacy, infrastructure scalability, and performance. While this shift enhances interoperability, CMS should manage the transition carefully to avoid penalizing smaller providers who may require time or resources to upgrade systems.

TD-10. For EHR and other developers subject to the ONC Health IT Certification Program, what further steps should ASTP/ONC consider to implement the 21st Century Cures Act's API condition of certification (42 U.S.C. 300jj-11(c)(5)(D)(iv)) that requires a developer's APIs to allow health information to be accessed, exchanged, and used without special effort, including providing access to all data elements of a patient's electronic health record to the extent permissible under applicable privacy laws?

To fulfill the API condition of certification under the 21st Century Cures Act, ONC should require certified developers to ensure complete, standards-based access to all electronic health record data elements without undue complexity or proprietary restrictions. This includes enforcing consistent implementation of FHIR APIs across vendors and expanding real-world testing to confirm usability and completeness of API responses. Additionally, ONC should promote modular certification pathways to reduce entry barriers for smaller developers while maintaining rigorous interoperability and transparency standards. This approach would enable broader participation in the digital health ecosystem and ensure that APIs meet the law's intent of enabling access, exchange, and use of health information without special effort.

TD-11. As of January 1, 2024, many health IT developers with products certified through the ONC Health IT Certification Program are required to include the capability to perform an electronic health information export or "EHI export" for a single patient as well as for patient populations (45 CFR 170.315(b)(10)). Such health IT developers are also required to publicly describe the format of the EHI export. Notably, how EHI export was accomplished was left entirely to the health IT developer. Now that this capability has been in production for over a year, CMS and ASTP/ONC seek input on the following: a. Should this capability be revised to specify standardized API requirements for EHI export? b. Are there specific workflow aspects that could be improved? c. Should CMS consider policy changes to support this capability's use?

CMS should recommend that ONC revise the EHI export capability to include standardized API requirements—such as support for FHIR Bulk Data Access—to ensure consistency and scalability across vendors. This would reduce variation in implementation and lower provider burden. To improve workflows, the export function should support automation, audit logging, and flexible scheduling aligned with clinical and reporting cycles. CMS should also embed EHI export functionality into critical clinical workflows, such as discharge planning and payer handoffs, to ensure timely and accurate data transfer when it matters most. CMS could further support adoption by linking policy incentives to demonstrated use of EHI export capabilities, particularly for care coordination and quality reporting use cases. Incorporating requirements or credits for demonstrated use of standardized EHI export in quality reporting and value-based purchasing programs, such as the Merit-based Incentive Payment System (MIPS) will encourage providers to actively use and maintain compliant export functions. Publishing vendor interoperability scorecards—

including EHI export performance—would promote transparency and accountability in the certification ecosystem.

4.0 Data Exchange

TD-12. Should CMS endorse non-CMS data sources and networks, and if so, what criteria or metrics should CMS consider?

CMS should endorse non-CMS data sources and networks that meet clearly defined interoperability criteria, including conformance to FHIR R4+ standards, robust identity assurance protocols, and measurable data quality benchmarks such as completeness, accuracy, and timeliness. Establishing a "CMS Trusted Data Partner" designation for networks that meet these standards would promote consistency, foster trust, and support scalable, high-fidelity data exchange. Endorsement should be based on transparent metrics and real-world performance to ensure meaningful participation and alignment with CMS program goals.

TD-13. What new opportunities and advancements could emerge with APIs providing access to the entirety of a patient's electronic health information (EHI)? a. What are the primary obstacles to this? b. What are the primary tradeoffs between USCDI and full EHI, especially given more flexible data processing capabilities today?

APIs that provide access to the entirety of a patient's electronic health information (EHI) create new opportunities for longitudinal care coordination, patient-centered decision support, and advanced analytics. However, primary obstacles include inconsistent data availability across EHR systems, lack of standardization for unstructured data, and concerns around privacy, consent, and data security.

A key tradeoff between USCDI and full EHI is the balance between standardized, interoperable data (USCDI) and broader, more flexible datasets that include narrative notes, scanned documents, and other unstructured elements. A transition to full EHI would provide a far broader range of clinical content—narrative notes, historical context, and diagnostic records—that can offer deeper insights into a patient's experience, albeit with greater complexity due to challenges around consistency, bias, and interpretability.

Nevertheless, the rise of natural language processing (NLP), machine learning, and improved cloud infrastructure make the processing of unstructured data more feasible than in the past. To manage this transition, CMS could implement an API "tiers of access" model, where access is granted based on application trust level, user consent, and adherence to data handling standards. In addition, ONC's Standard Advancement Process (SVAP) can help manage these tradeoffs by providing an evolving path for adopting advanced FHIR resources and profiles that bridge the gap between USCDI and full EHI,

including future inclusion of structured narrative elements or expanded document reference capabilities.

TD-14. Regarding networks' use of FHIR APIs: a. How many endpoints is your network connected to for patient data sharing? What types, categories, geographies of endpoints do you cover? Are they searchable by National Provider Identifier (NPI) or organizational ID? b. How are these connections established (for example, FHIR (g)(10) endpoints, TEFCA/Integrating the Health Enterprise (IHE) XCA, or proprietary APIs)? c. Do you interconnect with other networks? Under what frameworks (for example, TEFCA, private agreements)?

While we are not a health information network, it is clear from industry trends that leading networks now support connections to thousands of endpoints—ranging from hospitals, labs, and payers to community clinics and public health departments. These networks often span multiple states or regions. However, many current endpoint directories are fragmented and not reliably searchable by National Provider Identifier (NPI) or organizational ID, limiting discoverability. A standardized, NPI-linked directory API could improve searchability and help reduce provider burden.

Connections are commonly established through a mix of technologies. Many networks support FHIR (g)(10) APIs as required under the 21st Century Cures Act Final Rule, alongside legacy protocols like IHE XCA (Cross-Community Access). Proprietary APIs are still in use, but the strategic direction from both ONC and CMS is to prioritize open standards. ONC's Standard Version Advancement Process (SVAP) helps guide safe, incremental adoption of newer versions of FHIR, aligning with goals in HTI-1 through HTI-3.

TEFCA is emerging as the national standard for trusted data exchange across networks, aiming to provide secure, scalable interoperability. Until TEFCA is fully adopted, many rely on private agreements or data use arrangements to enable cross-network exchange. Standardized, proctored testing for network certification could further ensure consistent and unbiased implementation, reducing risks from loosely validated connections.

To promote network connectivity through FHIR APIs, CMS should follow a focused approach. Many networks lack searchable, centralized endpoint directories, requiring directories indexed by NPI or organization ID could support easier discovery and lower barriers for authorized access. FHIR (g)(10) APIs are not uniformly implemented, which leads to inconsistent experiences; SVAP could help modernize conformance expectations over time. TEFCA participation is growing, but broader alignment across CMS quality programs and payment models would accelerate adoption. CMS and ASTP/ONC should jointly promote reciprocity and transparency in network participation, reinforcing patient safety and reducing duplicative infrastructure.

This approach supports consistent, equitable access to patient data and aligns with both CMS payment reform and ASTP/ONC's interoperability goals under HHS. A focus on validated, standards-based APIs—tested under proctored conditions—ensures fairer results, fewer implementation failures, and safer, more meaningful exchange of health information.

TD-15. Regarding bulk FHIR APIs: a. How would increased use of bulk FHIR improve use cases and data flow? b. What are the potential disadvantages of their use?

Promoting Bulk FHIR APIs can significantly enhance data flows for population health management, quality reporting, and risk adjustment by enabling efficient, large-scale data extraction from EHR systems. This facilitates advanced analytics, supports research-grade datasets, and reduces manual reporting burdens on providers. However, disadvantages include the infrastructure demands on smaller organizations, concerns over data privacy at scale, and the potential for inconsistent implementation across vendors. CMS should incentivize Bulk FHIR adoption while also providing toolkits, shared services, or grants to help smaller networks manage the associated storage, bandwidth, and security requirements. CMS should also encourage developer accountability via formal testing requirements, reducing the compliance burden on implementers downstream.

TD-16. What are the tradeoffs of maintaining point-to-point models vs. shared network infrastructure? a. Do current rules encourage scalable network participation? b. What changes would improve alignment (for example, API unification, reciprocal access)?

CMS should transition away from point-to-point data exchange models, which are costly to maintain, difficult to scale, and prone to inconsistencies in access and interoperability. A shared network infrastructure—such as those aligned with TEFCA—can reduce redundancy, improve data quality, and support broader participation. Current rules do not yet fully incentivize scalable network models, particularly for smaller or rural providers. CMS should encourage API unification around FHIR R4+ standards, require reciprocal data access among participating entities, and offer phased access tiers based on identity assurance and data use. Enhancing financial and technical support for joining shared frameworks would further promote scalable participation.

TD-17. Given operational costs, what role should CMS or ASTP/ONC or both have in ensuring viability of healthcare data sharing networks, including enough supply and demand, that results in usage and outcomes?

To ensure the long-term viability of healthcare data sharing networks, CMS and ONC should play a proactive role in balancing supply and demand by fostering both participation and real-world utility. This includes offering grants or shared services contracts to help small and rural providers join networks, as well as incentivizing data contribution and

usage through value-based care programs. CMS should also promote robust privacy and consent frameworks—such as OAuth 2.0 and SMART on FHIR—to build patient trust and enable meaningful control over data access. By aligning financial support, technical standards, and regulatory incentives, CMS can drive network adoption and ensure these systems yield measurable outcomes in care quality, equity, and efficiency.

5.0 Compliance

TD-18. Information blocking: a. Could you, as a technology vendor, provide examples for the types of practices you have experienced that may constitute information blocking. Please include both situations of non-responsiveness as well as situations that may cause a failure or unusable response? b. What additional policies could ASTP/ONC and CMS implement to further discourage healthcare providers from engaging in information blocking practices? c. Are there specific categories of healthcare actors covered under the definition of information blocking in section 3022(a)(1) of the Public Health Service Act (PHSA) that lack information blocking disincentives?

Leidos has not directly encountered information blocking practices (in the context of this question) in the systems we develop or operate. However, we are aware that information blocking often manifests through non-responsiveness to API calls, the use of proprietary standards that hinder data portability, or imposing burdensome access conditions that render responses unusable. To further discourage such practices, CMS and ONC could implement more granular compliance reporting tied to specific interoperability metrics, increase transparency through public dashboards of API performance and availability, and establish anonymous whistleblower reporting channels. Additionally, CMS could extend disincentive frameworks to more consistently cover actors such as health information networks (HINs) and developers of health IT who may currently lack enforcement clarity under Section 3022(a)(1) of the PHSA. These actions would strengthen accountability while promoting a culture of open and equitable data sharing.

TD-19. Regarding price transparency implementation: a. What are current shortcomings in content, format, delivery, and timeliness? b. Which workflows would benefit most from functional price transparency? c. What improvements would be most valuable for patients, providers, or payers, including CMS? d. What would further motivate solution development?

While CMS's hospital and payer price transparency rules mark important progress toward empowering patients with cost information, several systemic challenges limit their effectiveness. The absence of a common data schema or machine-readable standard hinders consistent data ingestion, impairs automation, and creates unnecessary burden for third-party developers. Update frequency is inconsistent—ranging from monthly to infrequent refreshes—resulting in outdated data that erodes trust and utility. Additionally,

the technical nature of the information, such as reliance on unexplained CPT or HCPCS codes, makes it inaccessible to most consumers, especially for bundled services. Finally, the lack of standardized delivery channels limits discoverability and integration into clinical workflows, ultimately falling short of enabling timely, informed healthcare decisions.

Workflows that would benefit most from functional price transparency include patient financial counseling, clinical decision-making, prior authorization, and payer network design. Real-time access to pricing data allows clinicians and patients to choose care options that align with both medical needs and financial circumstances, improving shared decision-making. During prior authorization and referrals, transparent cost information supports faster, more informed approvals and reduces unexpected billing. For payers and CMS, access to structured price data enables more effective value-based plan design and provider network optimization. Additionally, public health workflows benefit through improved analysis of pricing disparities, supporting more equitable policy and resource decisions.

The most valuable improvements for patients, providers, and payers—including CMS—are those that enhance the usability, accuracy, and interoperability of price transparency data, making it actionable in real-world healthcare decisions. Key improvements include the development of a standardized FHIR® Implementation Guide specific to price transparency use cases, embedding SMART on FHIR APIs into patient-facing apps for real-time, personalized cost comparisons, and supporting real-time APIs that reflect current contracts and benefits. Certification programs should go beyond technical compliance to evaluate usability and data accuracy, while voluntary "Transparency Excellence Badges" could incentivize user-centered design and equitable access. For CMS, structured pricing data supports policy analysis, Medicare Plan Finder enhancements, and program integrity. To further motivate solution development, CMS and ASTP/ONC should align transparency metrics with quality incentives, promote public-private collaboration, expand infrastructure support for small providers, and safeguard equity through privacy-by-design requirements—strategies that together will drive innovation, encourage market competition, and ensure fair, broad adoption of transparency tools.